

ANVESHA — benchmark report

SIH26169 · coarse pointing, acquisition & tracking for mobile FSOC terminals · team Regnum Carya

software-in-the-loop simulation

ground-truth metrics

identical seeds for every pipeline

16 scenarios × 4 pipelines × 3 seeds = 192 runs · seeds 1, 2, 3 · benchmark run 2026-09-30 18:00 on Windows-11-10.0.26100-SP0 · 24 logical CPUs · Python 3.14.7

1 · Executive summary

90 %

runs acquired ≤ 2 s

baselines 10–15 %

1.23 s

median acquisition time

baselines 7.2–9.2 s

11/16

scenarios meeting every PS limit

baselines 0

42

slowest closed-loop FPS (PS ≥ 20)

16 scenarios

- ANVESHA acquired the beacon within 2 s in **90 %** of runs; the three spiral-search baselines managed 10–15 % on identical seeds.
- Median (over scenarios) acquisition time: **1.23 s** vs 7.2–9.2 s.
- Every SIH26169 limit was met simultaneously in **11/16** scenarios (baselines: 0, 0, 0).
- Slowest closed loop (simulation + rendering + tracking) ran at **42 FPS**, above the 20 FPS requirement in every scenario.
- **Not met:** F2 vibration (Acq, Err); J fast (Err); K occlusion (Err); N worst (Acq, Err, Loss); O distractors (Acq). See limitations.

2 · SIH26169 compliance matrix

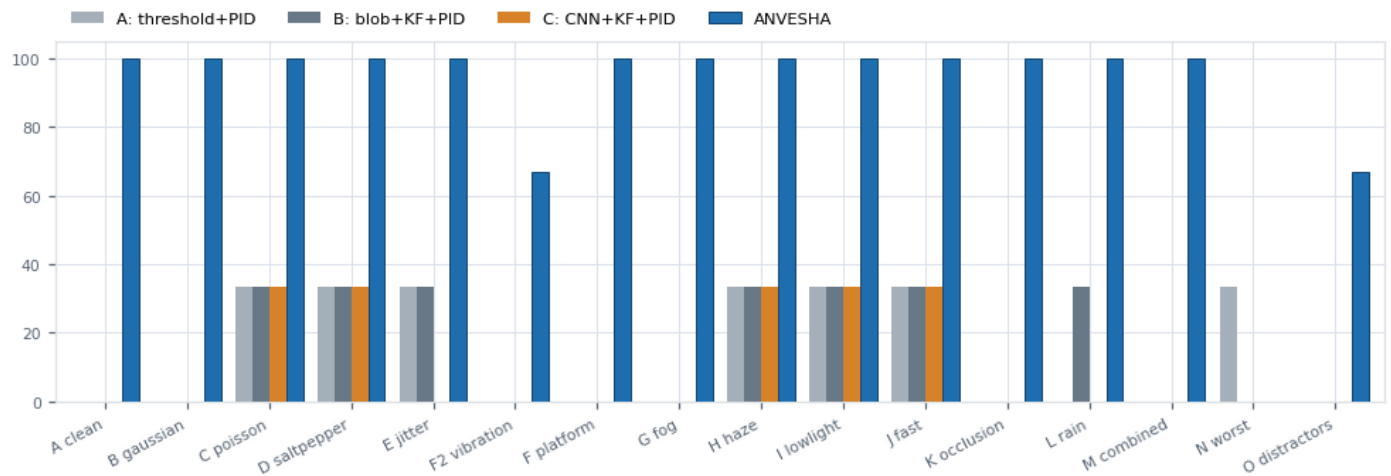
✓ = every limit met in the seed-aggregated result. Chips: **Acq** acquisition ≤ 2 s in every seed · **Err** mean tracking error ≤ 10 px (LOS and image) · **Loss** target loss < 5 % (excl. scripted occlusion) · **Reacq** re-acquisition ≤ 1 s, none unrecovered · **FPS** closed-loop ≥ 20 FPS. Grey chip = not applicable (no loss event).

Scenario	A: threshold+PID						B: blob+KF+PID						C: CNN+KF+PID						ANVESHA					
A clean	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✓	Acq	Err	Loss	Reacq	FPS
B gaussian	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✓	Acq	Err	Loss	Reacq	FPS
C poisson	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✓	Acq	Err	Loss	Reacq	FPS
D saltpepper	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✓	Acq	Err	Loss	Reacq	FPS
E jitter	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✓	Acq	Err	Loss	Reacq	FPS
F2 vibration	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS
F platform	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✓	Acq	Err	Loss	Reacq	FPS
G fog	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✓	Acq	Err	Loss	Reacq	FPS
H haze	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✓	Acq	Err	Loss	Reacq	FPS
I lowlight	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✓	Acq	Err	Loss	Reacq	FPS
J fast	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS
K occlusion	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS
L rain	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✓	Acq	Err	Loss	Reacq	FPS
M combined	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✓	Acq	Err	Loss	Reacq	FPS
N worst	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS
O distractors	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS	✗	Acq	Err	Loss	Reacq	FPS

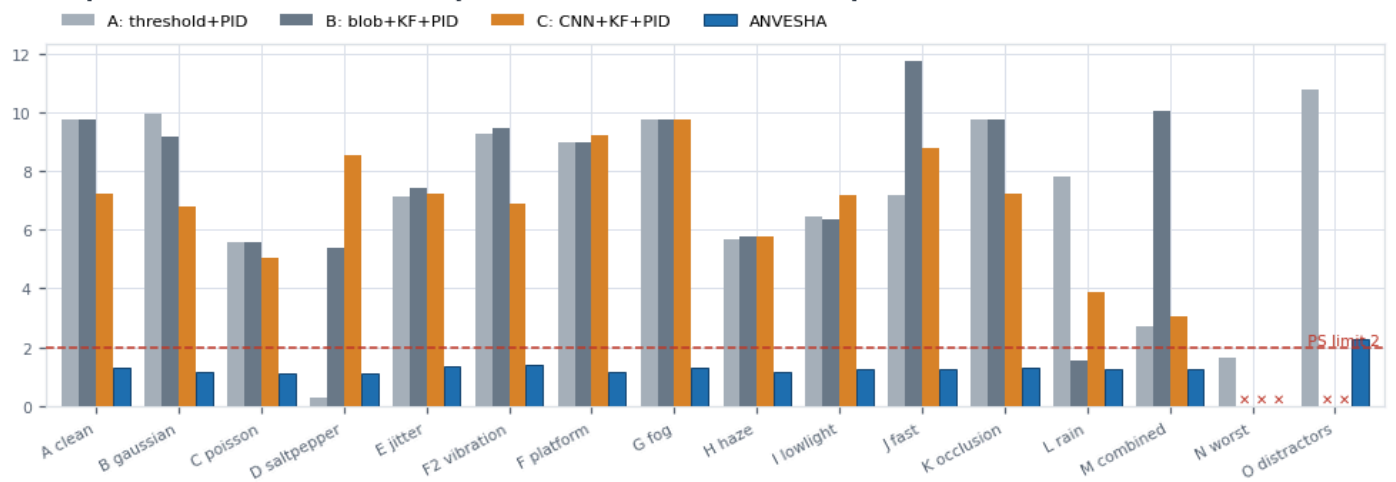
3 · Pipeline comparison

Bars are means over seeds. Red dashed line = PS limit. × = metric undefined because the pipeline never acquired; ▲ = value above the axis cap.

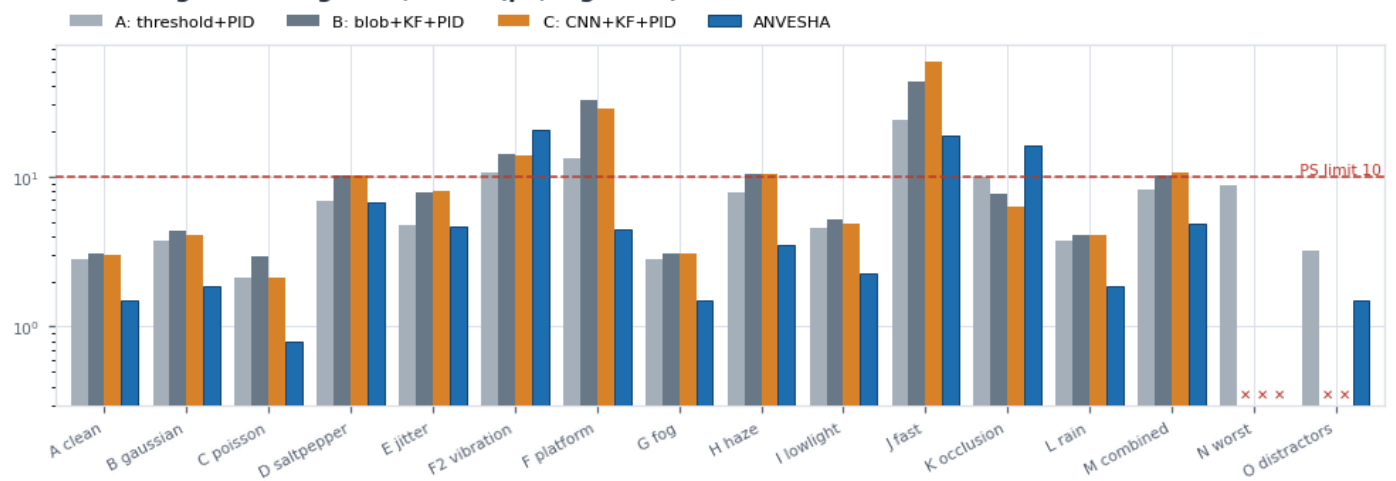
Runs acquired within 2 s (%) — higher is better

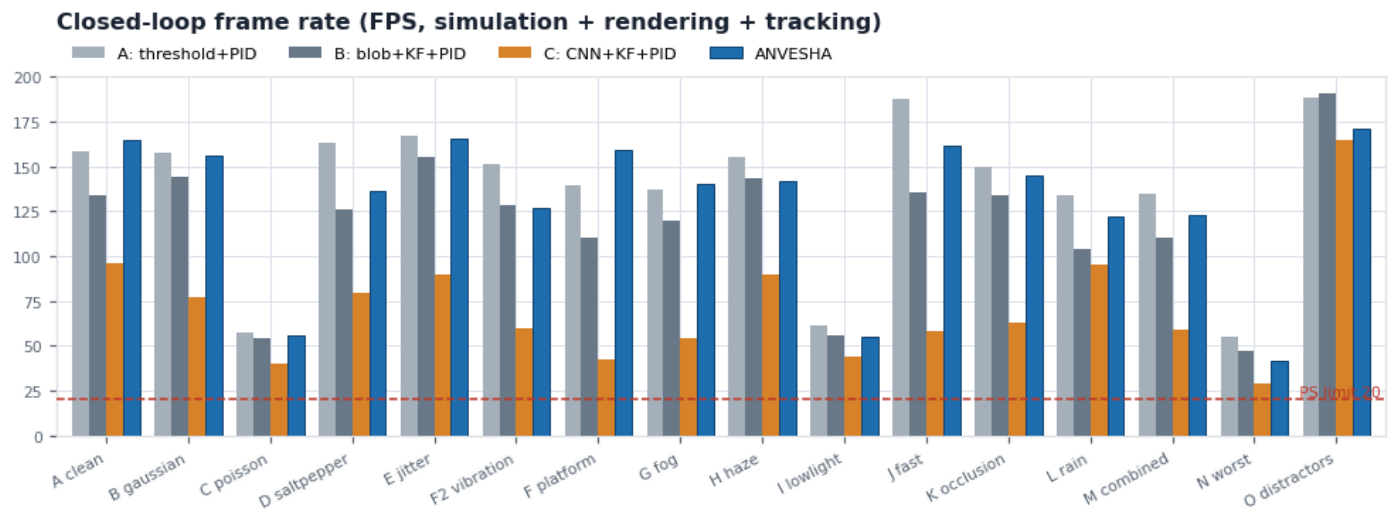
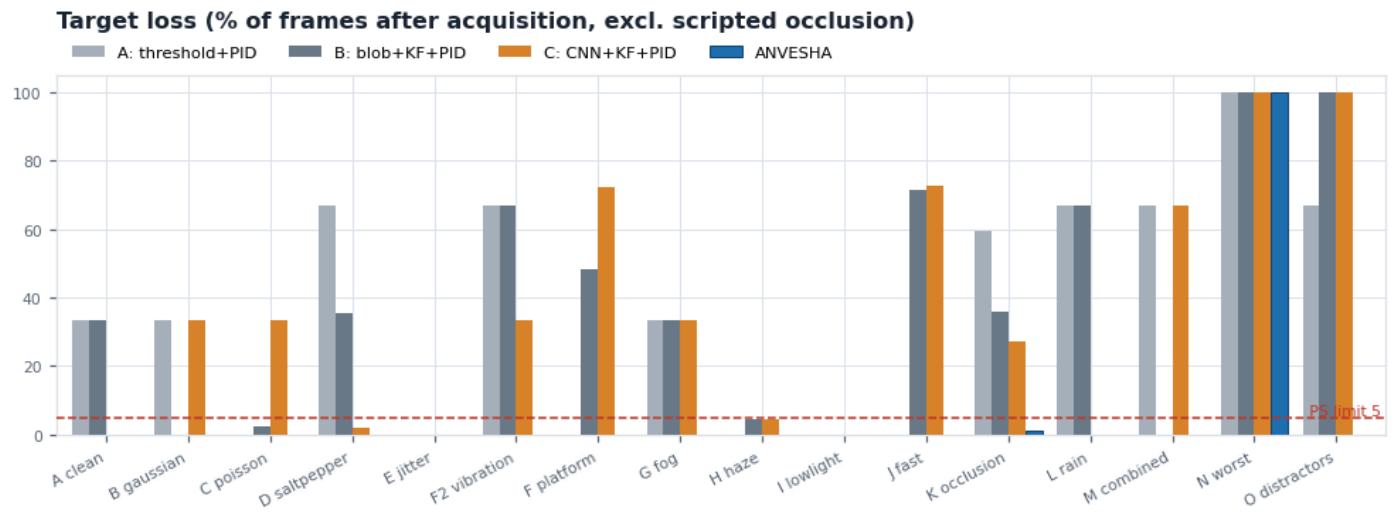


Acquisition time, mean over acquired runs (s) — × = never acquired



Line-of-sight tracking error, mean (px, log scale)





4 · Ablation — what each component buys

ANVESHA with exactly one component removed; 10 scenarios × seeds from `ablation/summary.json`. Red = worse than the full system.

Variant	Runs acquired ≤ 2 s (%)	Acq. time mean (s)	LOS error mean (px)	False-lock frames (sum)
ANVESHA (full)	87	1.36	6.7	173
– belief search (spiral)	7	7.80	8.5	105
– adaptive FOV	20	3.76	9.2	150
– IMU feed-forward	90	1.37	11.6	477
– jitter-aware R	80	3.07	12.6	291
– CNN verifier	83	1.49	6.7	219
IMM → KF-CV	87	1.89	10.5	132

- jitter-aware R: LOS error in E_jitter 4.6 → **19.3**
- jitter-aware R: LOS error in M_combined 4.9 → **18.4**
- IMU feed-forward: LOS error in F_platform 4.4 → **5.8**
- CNN verifier: acquisition time in O_distractors 2.3 → **3.4**

5 · Methodology & metric definitions

Simulation

240 Hz physics, 30 Hz 640×480 camera (4°×3° base FOV), 60 Hz control, pan/tilt limited to 5 °/s; seeded PCG64 streams per subsystem (bit-reproducible).

Truth isolation

The tracker receives only frames, encoder angles, IMU rates and timestamps; ground truth is used only by the metrics engine.

Baselines

A threshold+CoG+PID, B blob+KF+PID, C CNN+KF+PID; all use spiral search; PID gains grid-tuned (best set kp 12, ki 20, kd 0.05).

Acquisition time

Run start → first frame in TRACK with true LOS error ≤ 10 px. "Acq ≤ 2 s %" counts seeds meeting this.

LOS / image error

Distance between beacon and boresight after acquisition; LOS excludes the white camera-jitter component (uncorrectable by a coarse gimbal), image error includes it.

Target loss

% of frames after acquisition with LOS error > 10 px or no lock, excluding scripted occlusion windows.

Re-acquisition

Per loss episode, from loss (or end of scripted occlusion) to the next on-target frame; unrecovered episodes counted separately.

Centroiding error

Raw per-frame centroid vs true sub-pixel beacon position (camera px).

FPS

Closed-loop rate including scene rendering, tracking and control on the machine above.

6 • Full results

Green = PS limit met, red = not met. Means over seeds.

Scenario	Pipeline	Acq ≤2 s %	Acq mean s	Acq max s	LOS err px	Img err px	Centroid RMS px	Loss %	Re- acq max s	Unrecov.	False- lock fr	FPS
A clean	A: threshold+PID	0	9.78	10.47	2.80	2.80	0.112	33.3	—	0	0.0	159
A clean	B: blob+KF+PID	0	9.78	10.47	3.06	3.06	0.105	33.3	—	0	0.0	134
A clean	C: CNN+KF+PID	0	7.23	10.47	2.99	2.99	0.040	0.0	—	0	0.0	96
A clean	ANVESHA	100	1.29	1.60	1.48	1.48	0.010	0.0	—	0	0.0	164
B gaussian	A: threshold+PID	0	9.97	10.70	3.74	3.74	0.064	33.3	—	0	1.0	158
B gaussian	B: blob+KF+PID	0	9.17	9.47	4.31	4.31	0.077	0.0	—	0	0.0	145
B gaussian	C: CNN+KF+PID	0	6.80	10.37	4.04	4.04	0.070	33.3	—	0	1.7	78
B gaussian	ANVESHA	100	1.16	1.40	1.86	1.86	0.079	0.0	—	0	0.0	156
C poisson	A: threshold+PID	33	5.58	8.77	2.11	2.11	0.069	0.0	—	0	0.0	57
C poisson	B: blob+KF+PID	33	5.58	8.77	2.93	2.93	0.076	2.4	0.57	0	0.0	54
C poisson	C: CNN+KF+PID	33	5.07	9.77	2.10	2.10	0.063	33.3	—	0	80.3	40
C poisson	ANVESHA	100	1.10	1.33	0.79	0.79	0.090	0.0	—	0	0.0	56
D saltpepper	A: threshold+PID	33	0.30	0.30	6.90	6.90	0.198	66.7	—	0	400.0	163
D saltpepper	B: blob+KF+PID	33	5.37	10.43	10.10	10.10	0.107	35.3	0.57	0	0.0	126
D saltpepper	C: CNN+KF+PID	33	8.57	14.97	10.04	10.04	0.098	2.0	0.57	0	0.3	80
D saltpepper	ANVESHA	100	1.12	1.37	6.74	6.74	0.106	0.0	—	0	0.0	136
E jitter	A: threshold+PID	33	7.13	10.43	4.78	9.02	0.022	0.0	—	0	0.0	167
E jitter	B: blob+KF+PID	33	7.44	10.43	7.82	10.76	0.028	0.0	—	0	0.0	155
E jitter	C: CNN+KF+PID	0	7.23	10.43	8.00	10.93	0.012	0.0	—	0	0.0	90
E jitter	ANVESHA	100	1.34	1.60	4.63	8.89	0.010	0.0	—	0	0.0	165
F2 vibration	A: threshold+PID	0	9.27	9.27	10.65	10.65	0.090	66.7	—	0	0.0	151
F2 vibration	B: blob+KF+PID	0	9.47	9.47	14.00	14.00	0.116	66.7	—	0	0.0	129
F2 vibration	C: CNN+KF+PID	0	6.90	9.47	13.79	13.79	0.069	33.3	—	0	0.3	60
F2 vibration	ANVESHA	67	1.40	2.13	20.54	20.54	0.010	0.0	—	0	0.0	127

F platform	A: threshold+PID	0	8.97	10.37	13.28	13.28	0.010	0.0	—	0	0.0	139
F platform	B: blob+KF+PID	0	8.97	10.37	32.52	32.52	0.092	48.1	4.67	1	0.0	110
F platform	C: CNN+KF+PID	0	9.23	10.47	28.04	28.04	0.080	72.2	2.33	2	3.0	43
F platform	ANVESHA	100	1.14	1.30	4.42	4.42	0.010	0.0	—	0	0.0	159
G fog	A: threshold+PID	0	9.78	10.47	2.81	2.81	0.121	33.3	—	0	0.0	137
G fog	B: blob+KF+PID	0	9.78	10.47	3.06	3.06	0.127	33.3	—	0	0.0	120
G fog	C: CNN+KF+PID	0	9.78	10.47	3.08	3.08	0.088	33.3	—	0	0.0	54
G fog	ANVESHA	100	1.29	1.60	1.48	1.48	0.091	0.0	—	0	0.0	141
H haze	A: threshold+PID	33	5.68	9.37	7.76	7.76	0.025	0.0	—	0	0.0	155
H haze	B: blob+KF+PID	33	5.77	9.37	10.39	10.39	0.030	4.2	0.57	0	0.0	144
H haze	C: CNN+KF+PID	33	5.78	9.37	10.38	10.38	0.037	4.2	0.57	0	0.0	90
H haze	ANVESHA	100	1.16	1.40	3.52	3.52	0.027	0.0	—	0	0.0	142
I lowlight	A: threshold+PID	33	6.44	9.63	4.50	4.50	0.082	0.0	—	0	0.0	61
I lowlight	B: blob+KF+PID	33	6.38	9.70	5.13	5.13	0.087	0.0	—	0	0.0	56
I lowlight	C: CNN+KF+PID	33	7.20	10.37	4.81	4.81	0.087	0.0	—	0	0.3	44
I lowlight	ANVESHA	100	1.23	1.43	2.24	2.24	0.094	0.0	—	0	0.0	55
J fast	A: threshold+PID	33	7.19	11.13	23.54	23.54	0.010	0.0	—	0	0.0	188
J fast	B: blob+KF+PID	33	11.77	17.80	42.87	42.87	0.059	71.4	3.57	2	0.0	136
J fast	C: CNN+KF+PID	33	8.77	15.93	58.27	58.27	0.142	72.6	3.57	1	0.7	58
J fast	ANVESHA	100	1.27	1.53	18.60	18.60	0.010	0.0	—	0	0.0	162
K occlusion	A: threshold+PID	0	9.78	10.47	9.84	9.84	0.154	59.5	5.83	0	0.0	150
K occlusion	B: blob+KF+PID	0	9.78	10.47	7.61	7.61	0.111	35.9	0.40	0	0.0	134
K occlusion	C: CNN+KF+PID	0	7.23	10.47	6.27	6.27	0.041	27.3	0.40	1	0.0	63
K occlusion	ANVESHA	100	1.29	1.60	16.18	16.18	0.013	1.2	0.27	0	0.0	145
L rain	A: threshold+PID	0	7.80	7.80	3.74	3.74	0.063	66.7	—	0	475.0	134
L rain	B: blob+KF+PID	33	1.53	1.53	4.07	4.07	0.066	66.7	—	0	8.7	104
L rain	C: CNN+KF+PID	0	3.90	7.50	4.09	4.09	0.078	0.0	—	0	3.3	95
L rain	ANVESHA	100	1.23	1.40	1.86	1.86	0.115	0.0	—	0	0.0	122

M combined	A: threshold+PID	0	2.70	2.70	8.20	10.01	0.297	66.7	0.03	0	418.3	135
M combined	B: blob+KF+PID	0	10.03	11.50	10.18	11.68	0.128	0.0	—	0	0.0	110
M combined	C: CNN+KF+PID	0	3.03	3.03	10.50	11.90	0.062	66.7	—	0	23.0	59
M combined	ANVESHA	100	1.23	1.50	4.89	7.82	0.095	0.0	—	0	0.0	123
N worst	A: threshold+PID	33	1.63	1.63	8.81	27.65	6.024	99.9	—	1	595.7	55
N worst	B: blob+KF+PID	0	—	—	—	—	0.494	100.0	—	0	33.7	47
N worst	C: CNN+KF+PID	0	—	—	—	—	0.858	100.0	—	0	43.3	29
N worst	ANVESHA	0	—	—	—	—	0.551	100.0	—	0	150.3	42
O distractors	A: threshold+PID	0	10.80	10.80	3.18	3.18	0.013	66.7	—	0	424.7	189
O distractors	B: blob+KF+PID	0	—	—	—	—	—	100.0	—	0	590.3	191
O distractors	C: CNN+KF+PID	0	—	—	—	—	—	100.0	—	0	588.3	165
O distractors	ANVESHA	67	2.29	3.70	1.50	1.50	0.018	0.0	—	0	23.0	171

7 · Limitations & honest notes

- All results are from simulation; no real camera or gimbal was used. The learned verifier is trained and evaluated on synthetic patches only.
- Uniform ± 20 px white jitter alone gives ≈ 16.3 px radial RMS image error — above 10 px for any coarse gimbal; a fine-steering stage is required.
- Vibration above the gimbal bandwidth and very fast manoeuvres raise LOS error above 10 px; scripted occlusions add error while the beacon is invisible.
- The all-maxima stress case (every PS disturbance at its maximum plus fog and look-alikes) is not solved.